

Remark	Recurrence	Encoding	Loss	PPL init	PPL best	Attn Len
Transformer-XL (128M)	✓	Ours	Full	27.02	26.77	500
-	✓	Shaw et al. (2018)	Full	27.94	27.94	256
-	✓	Ours	Half	28.69	28.33	460
-	✗	Ours	Full	29.59	29.02	260
-	✗	Ours	Half	30.10	30.10	120
-	✗	Shaw et al. (2018)	Full	29.75	29.75	120
-	✗	Shaw et al. (2018)	Half	30.50	30.50	120
-	✗	Vaswani et al. (2017)	Half	30.97	30.97	120
Transformer (128M) [†]	✗	Al-Rfou et al. (2018)	Half	31.16	31.16	120
Transformer-XL (151M)	✓	Ours	Full	23.43	23.09 23.16 23.35	640 450 300

Table 6: Ablation study on WikiText-103. For the first two blocks, we use a slightly smaller model (128M parameters). [†] indicates that the corresponding row is reduced to the same setting as the Transformer network in (Al-Rfou et al., 2018), except that two auxiliary losses are not implemented in our experiments. “PPL init” refers to using the same length as training. “PPL best” indicates the perplexity obtained by using the optimal length. “Attn Len” is the shortest possible attention length during evaluation to achieve the corresponding result (PPL best). Increasing the attention length during evaluation improves performance only when our positional encoding is used. The “Transformer-XL (151M)” setting uses a standard parameter budget as previous work (Merity et al., 2018), where we observe a similar effect when increasing the attention length during evaluation.

Method	PPL
Ours	25.2
With Shaw et al. (2018) encodings	25.7
Without recurrence	27.1

Table 7: Ablation study on One Billion Word, a dataset without long-term dependency.

Model	$r = 0.1$	$r = 0.5$	$r = 1.0$
Transformer-XL 151M	900	800	700
QRNN	500	400	300
LSTM	400	300	200
Transformer-XL 128M	700	600	500
- use Shaw et al. (2018) encoding	400	400	300
- remove recurrence	300	300	300
Transformer	128	128	128

Table 8: Relative effective context length (RECL) comparison. See text for the definition of RECL and r . The first three models and the last four models are compared as two *model groups* when we calculate RECL (RECL is computed on a model group rather than a single model). Each group has the same parameter budget.

called *Relative Effective Context Length* (RECL). RECL is defined on a model group instead of a single model, and the gain of a long context is measure by the relative improvement over the *best* short context model. As such, the model group shares the same baseline to enable fair comparison. RECL also has a parameter r , which means constraining the comparison on top- r hard examples. See Appendix C for more details about RECL. As shown in Table 8, Transformer-XL manages to model dependency of 900 words long on av-

Attn Len	How much Al-Rfou et al. (2018) is slower
3,800	1,874x
2,800	1,409x
1,800	773x
800	363x

Table 9: Slowdown in terms of running time during evaluation. Evaluation is based on per-token time on one GPU.

erage with $r = 0.1$. The RECL of Transformer-XL is 80% and 450% longer than recurrent networks and Transformer respectively. Both the recurrence mechanism and our positional encodings contribute to a longer RECL. This further substantiates our argument that Transformer-XL is able to model longer-term dependency.

4.4 Generated Text

Trained only on WikiText-103 which is medium-sized, Transformer-XL is already able to generate relatively coherent articles with thousands of tokens without manual cherry picking, despite minor flaws. Please refer to Appendix E for samples.

4.5 Evaluation Speed

Finally, we compare the evaluation speed of our model with the vanilla Transformer model (Al-Rfou et al., 2018). As shown in Table 9, due to the state reuse scheme, Transformer-XL achieves an up to 1,874 times speedup during evaluation.